Music Piece Classification via Learned Embeddings

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***Abstract*—This paper addresses the problem of fine-grained music piece identification: given an 8-second audio snippet, classify it as belonging to one of 430 Bach chorales against a random baseline of 0.23% top-1 / 1.16% top-5. We trace a full methodological arc from hand-crafted musical features (key, pitch, note density) combined with constrained logistic regression, through frozen MERT transformer embeddings with linear and MLP classifiers, to fine-tuning the MERT backbone itself and applying metric learning with an ArcFace-style projection head. Frozen-embedding methods plateaued in the 38–45% top-1 range, with the best frozen result obtained using leak-corrected pitch and tempo augmentation, revealing that embedding geometry rather than classifier capacity is the binding constraint. Fine-tuning the top four MERT transformer layers breaks through this ceiling, achieving 70.2% top-1 and 88.1% top-5 accuracy — a 305× improvement over the random baseline. These results demonstrate that task-specific adaptation of pre-trained audio transformers is crucial for strong fine-grained within-corpus classification, with potential implications for music recommendation, plagiarism detection, and archival search systems.**

***Index Terms*—music information retrieval, audio classification, learned embeddings, MERT**

I. Introduction

The ability to identify a musical work from a short audio excerpt underpins a wide range of practical applications. Music recommendation systems rely on piece-level identity to surface related works; plagiarism detection in creative industries requires matching newly composed material against existing catalogues; and archival search over large digitized music collections demands robust snippet-to-piece retrieval. Yet the core technical challenge is formidable: a few seconds of audio must be correctly attributed to one of hundreds or thousands of distinct pieces, with limited training examples per class and no guarantee that the query audio shares recording provenance with the reference.

This paper studies the problem in a controlled but challenging setting: classifying 8-second audio snippets drawn from 430 Bach chorales, where each snippet must be matched to its source piece against a random baseline of $1/430 \approx 0.23\%$ for top-1 accuracy and $5/430 \approx 1.16\%$ for top-5 accuracy. The Bach chorale corpus is a natural testbed — the pieces are harmonically sophisticated, stylistically homogeneous, and entirely in the public domain — but its homogeneity also makes fine-grained discrimination difficult. Pieces share key signatures, similar rhythmic textures, and closely related harmonic progressions, so classifiers must extract subtle identity signals rather than broad stylistic cues.

Our approach evolved through several distinct phases. We began with hand-crafted musical features (key signature, time signature, average pitch, pitch range, note density) extracted via music21 and combined through a constrained logistic regression with simplex-projected weights, framing piece identification as a binary match/non-match problem. Recognizing the limits of manually designed features, we pivoted to MERT — a transformer pre-trained on large-scale music audio — whose frozen 768-dimensional embeddings encode rich acoustic representations without any task-specific supervision. We trained linear and MLP classifiers on these frozen embeddings, systematically searched over architectures, and applied pitch-and-tempo data augmentation to expand per-class training signal. Finally, we fine-tuned the top layers of the MERT backbone for piece identity, finding that adapting the embedding space itself — rather than building a more powerful classifier on top of it — was the key to achieving strong performance.

The remainder of this paper is organized as follows. Section II surveys relevant prior work in audio fingerprinting, music information retrieval, self-supervised audio modeling, and metric learning. Section III formalizes the two problem formulations we employed. Section IV describes the Bach chorale dataset, our audio rendering pipeline, and the data augmentation and leakage analysis. Section V details each method in the order it was developed. Section VI presents quantitative results. Sections VII–IX discuss findings, limitations, and directions for future work, and Section X concludes.

Contributions. This paper makes the following contributions:

- A systematic comparison of eight classification approaches on a 430-class music identification benchmark, from logistic regression on hand-crafted features to fine-tuned transformer embeddings.
- An empirical demonstration that frozen MERT embeddings plateau in the 38–45% top-1 range, identifying representation quality as the binding constraint.
- A fine-tuning protocol for MERT achieving 70.2% top-1 / 88.1% top-5 accuracy with a 4-layer unfreezing strategy.
- An analysis of data augmentation-induced leakage as a methodological pitfall in snippet-level train/test splitting.

II. Related Work

a) Audio Fingerprinting.: The foundational work on exact-match audio identification is Wang's industrial-strength fingerprinting algorithm [1], which underpins the Shazam service. Wang's approach extracts a sparse constellation of time-frequency landmarks from a spectrogram and indexes combinatorially hashed pairs of peaks into a database; at query time, candidate matches are verified by checking that the time-offset differences between probe and reference fingerprints are consistent. The result is a system that is noise-resistant, computationally efficient, and capable of identifying a few-second clip against a database of millions of tracks. Our setting differs from fingerprinting in a fundamental way: fingerprinting assumes the query is a degraded copy of a known recording, whereas our task requires matching a *newly synthesized* audio snippet to its source piece, a relationship that involves different instrument timbres, synthesis artifacts, and no shared recording provenance. Robustness to these acoustic differences demands learned representations rather than exact hash matching.

b) Music Information Retrieval and Cover Song Detection.: The broader MIR literature has long addressed the problem of identifying musically equivalent content across different performances. Ellis and Poliner [2] proposed a chroma-feature approach combined with dynamic-programming beat tracking for cover song detection, demonstrating that pitch-class profiles can bridge different recordings of the same piece. The music21 toolkit [3] provided the symbolic-score infrastructure on which our feature extraction pipeline is built, offering programmatic access to key signature, time signature, pitch, and rhythmic structure from standard music notation formats. McFee *et al.* [4] introduced librosa, the Python library we rely on for audio loading, pitch shifting, and time stretching during data augmentation. In contrast to cover song detection—which identifies the same *song* across diverse arrangements—our task is fine-grained piece identification within a homogeneous corpus (Bach chorales in a single synthesized timbre), making the discriminative challenge harder at the within-corpus level and eliminating the need to be robust to radical harmonic reharmonizations.

c) Self-Supervised Models for Music Understanding.: Recent work has demonstrated that transformer-based self-supervised learning can produce rich general-purpose representations for audio. Li *et al.* [5] introduced MERT (Music undERstanding model with large-scale self-supervised Training), a BERT-style encoder pre-trained on large music corpora using masked prediction with dual teacher signals: an RVQ-VAE acoustic teacher and a Constant-Q Transform musical teacher that provides pitch and harmonic inductive bias. MERT scales from 95M to 330M parameters and achieves state-of-the-art performance across 14 downstream MIR tasks. We use the frozen MERT-95M encoder as a feature extractor in our initial experiments and subsequently fine-tune its top transformer layers for piece identification, making MERT both the representational backbone and a fine-tuning target of our work.

d) Metric Learning in Audio and Speaker Recognition.: Metric learning has been successfully applied to the related problem of speaker verification, where the goal is to learn an embedding space in which same-speaker utterances cluster tightly and different-speaker utterances are well-separated. Coria *et al.* [7] provide a systematic comparison of metric learning loss functions on the VoxCeleb dataset, finding that the additive angular margin loss outperforms contrastive and triplet objectives on speaker verification. Deng *et al.* [6] originally proposed ArcFace for face recognition: by adding a fixed angular margin m to the cosine angle between a feature and its class centre before the softmax, ArcFace maximizes the geodesic distance between classes on the hypersphere, yielding highly discriminative embeddings. We adopt an ArcFace-style cosine classifier as the final stage of our metric learning variant, leveraging the analogy between speaker and piece identity: both tasks require discriminating among a large number of classes from short, acoustically variable snippets.

e) Data Augmentation for Audio.: Park *et al.* [8] introduced SpecAugment, a simple policy of randomly masking contiguous time steps and frequency bins in the spectrogram, which achieves state-of-the-art results on speech recognition benchmarks by preventing over-reliance on any single spectro-temporal region. In the music domain, pitch shifting and tempo stretching are well-established waveform-level augmentations for enriching limited labeled data [4], [9]. Our augmentation strategy applies cross-product combinations of pitch shifts ($\pm 1, \pm 2$ semitones) and tempo factors ($\{0.8, 0.9, 1.0, 1.1, 1.2\}$) via librosa, yielding 25 variants per original snippet and expanding the training set from roughly 2,500 to 10,750 audio files. This strategy mirrors the robustness desiderata of cover song detection—a model should recognize a piece regardless of transposition or tempo—while remaining grounded in the perturbed-copy setting rather than the full cover song scenario.

III. Problem Formulation

This section documents how the problem formulation evolved over the course of the project — a progression that mirrors a genuine shift in what the task demanded, not merely a refinement of the same objective.

A. Original Formulation: Binary Similarity Classification

The initial formulation treated piece identification as a *binary similarity* problem: given two musical snippets, determine whether they originate from the same piece. Five features were extracted from the symbolic score via music21 — key signature, time signature, average pitch of the soprano voice, pitch range, and note density — and for each feature i, a heuristic similarity function was applied to produce a scalar $s_i \in [0, 1]$. The full similarity vector is $s = [s_1, s_2, s_3, s_4, s_5]^\top$.

a) Linear formulation.: The initial model computed a weighted sum $\hat{y} = w^\top s$ and minimized squared error against a binary label $y \in \{0, 1\}$ (1 for same-piece pairs, 0 otherwise), subject to simplex constraints on the weight vector:

$$\min_w \frac{1}{n} \sum_{i=1}^{n} (y_i - w^\top s_i)^2 \quad \text{subject to} \quad \sum_{j=1}^{5} w_j = 1, \; w_j \geq 0. \tag{1}$$

This problem was solved with three methods: Projected Gradient Descent (PyTorch), SLSQP (`scipy.optimize.minimize`), and convex programming (CVXPY). All three converged to nearly identical solutions with key signature and pitch range as the dominant features.

b) Logistic formulation.: Because (1) is a regression loss applied to a classification problem, the formulation was subsequently revised to binary logistic regression. The similarity scores were re-scaled to $[-1, 1]$ and the simplex equality constraint was relaxed to a non-negativity constraint ($w_j \geq 0$), allowing each feature's contribution to vary freely in magnitude. The revised objective is:

$$\begin{aligned} \min_{w} \quad & -\frac{1}{n}\sum_{i=1}^{n}\big[y_i \log \sigma(w^\top s_i) + (1-y_i)\log(1-\sigma(w^\top s_i))\big] \\ \text{s.t.} \quad & w_j \geq 0, \end{aligned} \tag{2}$$

where $\sigma(\cdot)$ is the sigmoid function. A pair is predicted as a match when $w^\top s_i \geq 0$. The same three solvers were applied; on an expanded dataset of 50 labeled pairs (Bach, Mozart, and Beethoven works), all three achieved $\approx 53.3\%$ test accuracy against a 50% random baseline.

c) Limitations.: The binary formulation, while tractable, has an inherent ceiling: it cannot identify *which* piece a snippet belongs to, only whether two snippets share a source. As the dataset grew and richer representations became available, this coarseness made it impossible to exploit the full 430-class structure of the Bach chorale corpus.

B. Revised Formulation: Multi-Class Piece Identification

The shift to MERT embeddings made a more ambitious formulation feasible. Rather than comparing pairs, the revised task is to classify each snippet directly into one of $C = 430$ Bach chorales.

a) Input representation.: Each audio snippet is passed through MERT-v1-95M, a transformer pre-trained on large-scale audio, and the hidden states are mean-pooled over the time dimension to yield a fixed-length embedding $x \in \mathbb{R}^{768}$.

b) Objective.: A weight matrix $W \in \mathbb{R}^{C\times d}$ and bias $b \in \mathbb{R}^C$ are learned to minimize cross-entropy over the training set:

$$\mathcal{L}(W, b) = -\frac{1}{n}\sum_{i=1}^{n}\log\frac{e^{z_{y_i}}}{\sum_{c=1}^{C} e^{z_c}}, \quad z = Wx + b \in \mathbb{R}^C. \tag{3}$$

The predicted piece is $\hat{y} = \arg\max_c z_c$.

c) Evaluation.: Performance is measured by top-k accuracy on a held-out test set of 521 snippets. The random baseline is $1/430 \approx 0.23\%$ for top-1 accuracy and $5/430 \approx 1.16\%$ for top-5 accuracy, so any method that materially exceeds these baselines confirms that the features contain piece-identity signal. Top-5 accuracy is reported alongside top-1 to characterize the shape of the score distribution. This evaluation measures generalization to held-out snippets from the same set of 430 known pieces; it does not test recognition of entirely unseen pieces.

The multi-class formulation is strictly more informative than the binary one: it diagnoses not only whether two snippets are from the same piece, but *which* piece each snippet is from. It also opens the door to per-class analysis that reveals, for instance, which chorales share confusable key signatures.

IV. Data

A. Dataset and Preprocessing

All musical material is drawn from the Bach chorale corpus available in the music21 library [3] — a collection of 430 four-part harmonizations catalogued under the Bach-Werke-Verzeichnis (BWV) numbering system. The corpus spans a wide range of keys and time signatures while remaining stylistically homogeneous: every piece is a short, fully harmonized Protestant chorale for soprano, alto, tenor, and bass voices. Because the works pre-date copyright by more than two centuries, the dataset raises no legal or ethical concerns around data access or redistribution.

The data pipeline proceeds in four stages. First, each chorale is loaded as a music21 `Score` object. Second, the score is rendered to a WAV audio file via FluidSynth, producing a synthetic but tonally faithful piano-like rendering. Third, each WAV is segmented into 8-second snippets with a 4-second overlap (later revised to a 2-second overlap after the data-leakage incident described in Section IV-C), generating multiple training examples per piece. Fourth, each snippet is passed through MERT-v1-95M and the resulting 768-dimensional mean-pooled embedding is cached to disk, so that downstream classifiers never need to re-run the full encoder.

The final unperturbed dataset contains **2,487 training snippets** and **521 test snippets** across all 430 pieces, with per-class counts ranging from roughly 3 to 40 snippets depending on chorale length. The train/test split is performed at the snippet level, stratified by piece, so that every piece's embedding distribution is represented in both splits. Because snippets are extracted with overlap, nearby windows from the same chorale may share audio content across the train/test boundary. This means the evaluation should be interpreted as held-out snippet classification within known pieces, rather than a fully independent piece-level generalization test.

B. Data Augmentation via Perturbation

A core challenge in piece identification is robustness: a human listener recognizes a chorale even when it is played slightly faster or transposed by a semitone, and a practical classifier should do the same. To encourage this invariance, the audio pipeline was extended with two families of perturbation implemented via the librosa library [4].

Pitch shifting transposes each WAV by $\{-2, -1, 0, +1, +2\}$ semitones. *Tempo stretching* rescales the playback speed by factors $\{0.8, 0.9, 1.0, 1.1, 1.2\}$ while preserving pitch. Combining all five pitch shifts with all five tempo factors yields **25 variants per original snippet**, expanding the unperturbed corpus of approximately 2,487 training snippets to **77,070 total WAV files** across train and test. In addition to mitigating memorization of clean audio, this augmentation deliberately mirrors the cover-song identification scenario: a perturbed snippet represents the same musical identity expressed under a modest transformation, exactly the challenge faced by real-world music-recognition systems.

C. Data Leakage and Revised Splitting

An early version of the perturbation pipeline split WAV files into train and test *after* perturbation, without tracking which snippets shared an original source. As a result, a snippet and several of its pitch/tempo variants could end up on opposite sides of the split. Because perturbed versions of the same snippet are nearly identical in embedding space, classifiers trained on this split achieved artificially inflated test accuracies — the MLP search reported top-1 accuracy of 87.8% under this condition, compared to 44.9% on the correctly split data.

The corrected pipeline performs splitting at the *snippet* level before perturbation: all 25 variants of a given snippet are assigned exclusively to either the training set or the test set. This ensures that a model evaluated on a perturbed clip has never observed any variant of that clip during training, producing a clean measure of generalization.

This incident is worth documenting as a methodological observation. The leakage condition incidentally defines a meaningful experimental regime: it tests whether a model can identify a piece despite small pitch and tempo shifts, which is precisely the robustness property one would want in a music fingerprinting system. The clean condition, by contrast, tests generalization to held-out snippets from *pieces* already seen during training. Both are valid tasks; conflating them obscures which challenge is actually being solved.

V. Methods

The methods are described in chronological order to preserve the narrative of increasing model capacity. Each subsection covers the motivation for trying that approach at that stage, the core mathematical formulation, and the implementation details.

A. Hand-Crafted Feature Extraction

a) Motivation.: At the project's outset, symbolic score data was readily available through music21 and audio processing resources were limited. Designing features directly from the score was the most natural starting point, and it constrained the problem to a small, interpretable parameter space.

b) Formulation.: Five features were extracted from each snippet's music21 `Score` object: (1) *key signature* (tonic and mode), (2) *time signature*, (3) *average pitch of the soprano voice*, (4) *pitch range* (difference between highest and lowest note), and (5) *note density* (notes per beat). For each feature i, a hand-crafted similarity function f_i mapped a pair of snippets to a scalar $s_i \in [-1, 1]$, where $+1$ denotes maximal similarity and -1 denotes maximal dissimilarity. The full similarity vector for a snippet pair is $s = [s_1, s_2, s_3, s_4, s_5]^\top$.

c) Implementation.: Feature extraction was implemented in Python using music21's score-parsing API. Key was represented by tonic and mode; time signature similarity penalized metric differences linearly; pitch statistics were computed over the soprano part only. Similarity functions were defined heuristically and not learned.

B. Logistic Regression on Hand-Crafted Features

a) Motivation.: With the five-dimensional similarity vector in hand, the immediate question was how to weight the features optimally. The binary classification framing made logistic regression a natural fit. Three solvers were compared to validate that the optimization was finding the same minimum rather than being solver-dependent.

b) Formulation.: The model minimizes the binary cross-entropy loss

$$\min_w \; -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log \sigma(w^\top s_i) + (1-y_i)\log(1-\sigma(w^\top s_i))\right] \quad \text{s.t.} \quad w_j \geq 0, \tag{4}$$

where $y_i \in \{0, 1\}$ is the same-piece label and σ is the logistic sigmoid. The non-negativity constraint ($w_j \geq 0$) enforces that no feature can contribute negatively to the similarity score; it replaced the stricter simplex constraint $\sum_j w_j = 1$ used in the earliest experiments after the equality constraint was found to substantially reduce performance.

c) Implementation.: Three solvers were benchmarked: (1) *Projected Gradient Descent* (PGD) implemented in PyTorch, using `binary_cross_entropy_with_logits` and projection onto the non-negative orthant after each gradient step; (2) *SLSQP* via `scipy.optimize.minimize` with the inequality constraint $w_j \geq 0$; and (3) *CVXPY* treating the problem as a convex program. All three converged to weight vectors dominated by pitch range and key signature, achieving $\approx$ 53.3% binary test accuracy on a 50-example dataset, compared to a 50% random baseline.

C. MERT Embeddings

a) Motivation.: The five hand-crafted features were insufficient to distinguish 430 chorales: key and pitch alone cannot resolve the many pieces sharing the same tonal center. Rather than expanding the feature set manually, the project pivoted to MERT — a transformer encoder pre-trained on large-scale unlabelled audio — to obtain representations that encode rich acoustic and harmonic information without requiring expert feature engineering.

b) Formulation.: MERT-v1-95M [5] is a 95-million-parameter transformer trained on a mixture of music and general audio using masked acoustic modeling. Given an 8-second WAV file, MERT produces a sequence of frame-level hidden states $H \in \mathbb{R}^{T\times 768}$, where T is the number of time frames (approximately 400 for an 8-second clip at the model's default frame rate). The embedding used for all downstream classifiers is the *mean-pooled* representation:

$$x = \frac{1}{T}\sum_{t=1}^{T} h_t \in \mathbb{R}^{768}. \tag{5}$$

c) Implementation.: The full embedding pipeline proceeds as follows. Symbolic scores from music21 are rendered to 22 kHz mono WAV files via FluidSynth. Each WAV is fed through MERT with the `transformers` library using the `m-a-p/MERT-v1-95M` checkpoint. The mean-pooled hidden state from the final layer is saved to disk as a

NumPy array. Embeddings are computed once and cached; at approximately 400 frames $\times$ 2,487 snippets $\times$ 768 dimensions $\times$ 4 bytes, the full embedding matrix occupies $\approx$3 GB of CPU RAM. An important implementation finding: applying L2 normalization to the embeddings before the classifier dropped top-1 accuracy from 34.4% to 4.4%, indicating that MERT embedding magnitudes carry piece-identity information that unit-normalization discards.

D. Logistic Regression on MERT Embeddings

a) Motivation.: With 768-dimensional MERT embeddings available, the first question was how much piece-identity information a *linear* classifier could extract. This establishes a capacity baseline and tests whether the embeddings are linearly organized around piece identity.

b) Formulation.: The linear model learns $W \in \mathbb{R}^{C\times d}$ and $b \in \mathbb{R}^{C}$ to minimize the unregularized cross-entropy loss (3), a smooth unconstrained problem in 330,670 parameters ($768 \times 430 + 430$).

c) Implementation.: Adam (lr = 10^{-3}, weight_decay = 0, batch size 256) was used over 1,000 epochs. Adam was preferred over SGD because the 430-class loss landscape has poor conditioning; a comparison run using SGD with L2-normalized embeddings achieved only 4.4% top-1, confirming the optimizer and normalization choices. Weight decay was swept over $\{10^{-4}, 10^{-6}, 0\}$ with no meaningful change in test accuracy ($< 0.6\%$), indicating that performance is determined by capacity rather than overfitting. The model achieved **34.4% top-1 and 56.4% top-5** test accuracy — a 150$\times$ improvement over the 0.23% random baseline — while training accuracy saturated at 99.8%, revealing a clear capacity ceiling.

E. MLP Architecture Search

a) Motivation.: The linear classifier's generalization gap (99.8% train, 34.4% test) persisted even with weight decay, ruling out overfitting as the bottleneck. The diagnosis was that a linear boundary in $\mathbb{R}^{768}$ cannot separate 430 classes whose embeddings are not linearly arranged. A multi-layer perceptron adds the nonlinearity needed to carve more complex decision boundaries, and a systematic architecture search identifies the most effective configuration.

b) Formulation.: The flexible MLP maps $x \in \mathbb{R}^{768}$ through one or more hidden layers with batch normalization, ReLU activations, and dropout before a final linear projection to $C = 430$ logits:

$$x \xrightarrow{\text{Linear}} H_1 \xrightarrow{\text{[BN] ReLU Drop}} \cdots \xrightarrow{\text{Linear}} z \in \mathbb{R}^{430}. \qquad (6)$$

Cross-entropy with inverse-frequency class weighting and label smoothing is minimized to address the 3–40 snippet class imbalance.

c) Implementation.: Ten architecture configurations were evaluated, spanning hidden dimensions $\{[512], [1024], [2048], [1024, 512], [2048, 512], [512, 256, 128], [1024, 512, 256], [2048, 1024, 512]\}$ and dropout rates $\{0.2, 0.3, 0.4, 0.5\}$. Each configuration was trained for 300 epochs using Adam with cosine annealing (an early stopping proxy), and the top candidate was retrained for 1,000 epochs. Kaiming uniform initialization was used throughout. On the unperturbed dataset, shallow networks (one hidden layer, $H \leq 2048$) consistently outperformed deeper ones. The best unperturbed configuration reached **38.4% top-1 and 59.1% top-5** accuracy. After correcting the perturbation split to avoid leakage, the best augmented MLP reached **44.9% top-1 and 69.3% top-5** accuracy. In both cases, performance plateaued far below the fine-tuning results described in Section V-G.

F. K-Nearest Neighbors Baseline

a) Motivation.: KNN is a non-parametric probe of embedding geometry: if same-piece embeddings cluster tightly in the MERT space, even a simple nearest-neighbor rule should achieve high accuracy. Its result therefore serves as a direct diagnostic of whether the frozen embedding space is geometrically organized around piece identity.

b) Formulation.: Given a query embedding x, the predicted piece is the plurality class among the k nearest neighbors in the training set under cosine similarity.

c) Implementation.: `KNeighborsClassifier` from scikit-learn was used with `metric='cosine'`, sweeping $k \in \{1, 2, \ldots, 49\}$ to identify the optimal neighborhood size. The best top-1 KNN result was **20.4%** at $k = 1$, while the best top-5 result was **33.8%** at $k = 5$. Both results substantially underperformed the parametric classifiers, indicating that the frozen MERT embedding space is not naturally clustered by piece identity. The large gap between the linear classifier (which uses the full weight matrix to carve decision boundaries) and KNN (which relies on local geometry) further supports the conclusion that the embedding space is not natively organized for piece identity.

G. Fine-Tuning MERT

a) Motivation.: The 38–45% plateau across frozen-embedding methods — regardless of classifier depth — pointed to the embedding geometry itself as the limiting factor. Fine-tuning the top layers of MERT directly adapts the representation to the piece-identity task. A naive end-to-end fine-tune from random head initialization, however, risks catastrophic forgetting: a random classification head sends maximally noisy gradients into the transformer from the very first step.

b) Formulation.: Fine-tuning proceeds in two phases. *Phase 1* holds all MERT weights frozen and trains only the classification head for 20 epochs on the pre-cached embeddings. This produces a head that generates a meaningful loss signal before any transformer weights are modified. *Phase 2* unfreezes the top N MERT transformer layers and trains end-to-end for 30 additional epochs using differential learning rates: $\eta_{\text{transformer}} = 10^{-5}$ for the unfrozen layers and $\eta_{\text{head}} = 10^{-3}$ for the classification head. The smaller learning rate for the transformer prevents large updates that would destroy pre-trained representations.

c) Caching optimization.: Because only the top N layers are trainable, the outputs of the frozen bottom $12 - N$ layers are identical at every epoch. These intermediate activations are precomputed once before Phase 2 begins and cached in CPU RAM (approximately 3 GB at ≈ 400 frames $\times$ 2,487 snippets $\times$ 768 dims $\times$ 4 bytes), reducing per-batch MERT compute from 12 transformer layers to N.

d) Implementation.: The classification head architecture was fixed to the best configuration from the MLP search ($768 \rightarrow$ BN $\rightarrow 512 \rightarrow$ ReLU $\rightarrow$ Dropout$(0.3) \rightarrow 430$). Two Phase-2 configurations were evaluated: $N = 2$ and $N = 4$ layers unfrozen. Both achieved roughly 30.5% top-1 at the end of Phase 1, consistent with prior frozen-embedding results. After Phase 2, the 2-layer run peaked at **68.9% top-1 / 87.9% top-5** (epoch 29) and the 4-layer run reached **70.2% top-1 / 88.1% top-5** (epoch 30), with test loss still declining — suggesting the model had not fully converged within the compute budget. Neither run exhibited catastrophic forgetting, validating the two-phase warm-up strategy.

H. Metric Learning

a) Motivation.: Fine-tuning showed that modifying the embedding geometry substantially outperforms training only a classification head. Metric learning pursues the same goal from a different angle: rather than adapting the MERT layers directly, it trains a *projection head* to transform frozen MERT embeddings into a new space where same-piece snippets cluster more tightly and cross-piece distances are maximized. This approach does not require back-propagating through MERT and is thus less compute-intensive.

b) Formulation.: The model maps the mean-pooled MERT embedding $x \in \mathbb{R}^{768}$ through a projection network to a normalized embedding $\hat{h}$, which is then classified by a geometry-aware head. Three variants were implemented:

Standard softmax: The projected embedding is passed through a linear head and softmax.

Cosine classifier: Both the embedding and each class prototype vector $\hat{w}_c$ are L2-normalized. The logit for class c is a scaled cosine similarity:

$$z_c = s \cdot \hat{w}_c^\top \hat{h}, \tag{7}$$

where $s > 0$ is a learnable temperature scale.

ArcFace classifier [6]: An angular margin m is added to the target class angle at training time. Let $\theta_c = \arccos(\hat{w}_c^\top \hat{h})$. The target-class logit becomes

$$z_y = s \cdot \cos(\theta_y + m), \tag{8}$$

while all non-target logits use the standard cosine score $z_c = s \cos(\theta_c)$. The margin penalizes embeddings that are not sufficiently close to their correct class prototype, encouraging compact within-class clusters and larger angular separation between classes.

c) Implementation.: All three variants were trained on frozen MERT embeddings using Adam with cross-entropy loss and the same training schedule as the MLP search. The ArcFace variant achieved approximately **40% top-1 / 63% top-5**, outperforming earlier frozen-embedding classifiers but falling short of fine-tuning by roughly 30 percentage points. Full ablations across margin and scale hyperparameters were not completed within the project's compute budget, leaving the precise performance ceiling of the metric-learning approach as an open question.

VI. Results

Table I summarizes test-set performance for all methods on the 521-snippet held-out set (430-class classification). Against random baselines of 0.23% top-1 and 1.16% top-5 accuracy, every learned approach offers substantial gains; the best result of **70.2%** top-1 / **88.1%** top-5 was achieved by fine-tuning four transformer layers of MERT end-to-end.

a) Hand-Crafted Features.: The earliest experiments used logistic regression over a five-dimensional feature vector derived from the symbolic score (key, time signature, average soprano pitch, pitch range, and note density). Because the original problem was framed as binary same-piece classification rather than 430-class identification, a direct top-k comparison with subsequent methods is not meaningful; results are therefore reported separately as binary test accuracy. Because this preliminary experiment used a different binary same-piece task and a much smaller dataset, it serves as a historical baseline rather than a directly comparable method. All three solvers (PGD, SLSQP, CVXPY) converged to the same solution, achieving 53.3% binary test accuracy — barely above the 50% coin-flip baseline on a highly imbalanced test set. The near-zero loss decrease from initialization ($0.6930 \rightarrow 0.6902$) and the fact that only pitch range and key received non-zero weights indicate that the five hand-crafted features carry insufficient discriminative information for fine-grained piece identification.

b) Frozen MERT Embeddings.: Switching to mean-pooled MERT embeddings (768-dimensional) and a multi-class formulation produced an immediate step change. A linear classifier trained on frozen embeddings reached 34.4% top-1 / 56.4% top-5, a roughly $150\times$ improvement over the random top-1 baseline despite severe per-class sparsity (410 of 430 test classes predicted at 0% accuracy). An MLP architecture search (single hidden layer of size 512, dropout 0.3) improved modestly to 38.4% top-1 / 59.1% top-5 on the unperturbed dataset. Notably, deeper multi-layer architectures degraded performance, with a three-hidden-layer network falling back to approximately 27% top-1, suggesting that the bottleneck lies in the embedding geometry rather than classifier capacity. The best top-1 KNN result was 20.4% at $k = 1$, while the best top-5 result was 33.8% at $k = 5$. Both KNN results substantially underperformed the parametric classifiers, confirming that frozen MERT representations are not well-clustered by piece identity in the raw embedding space.

c) Data Augmentation and the Leakage Incident.: To address per-class data scarcity, the training set was augmented by applying 5 tempo factors $\times$ 5 pitch shifts, generating 25 variants per original snippet. An initial MLP search on this augmented set produced an apparent top-1 accuracy of 87.8% (top-5: 96.6%), but post-hoc inspection revealed a critical data-leakage error: perturbed variants of the same source snippet had been distributed across both sides of the train/test split

TABLE I
SUMMARY OF ALL METHODS. TOP-K ACCURACIES ON HELD-OUT TEST SET (521 SNIPPETS, 430-CLASS CLASSIFICATION). RANDOM BASELINE = 0.23% TOP-1 / 1.16% TOP-5. † DENOTES RESULTS AFFECTED BY DATA LEAKAGE; SEE SECTION IV-C.

Method	**Input**	**Top-1 (%)**	**Top-5 (%)**
Random baseline	—	0.23	1.16
Log. Reg. (PGD)	Hand-crafted (5-d)	53.3% binary acc.	
Log. Reg. (SLSQP)	Hand-crafted (5-d)	53.3% binary acc.	
Log. Reg. (CVXPY)	Hand-crafted (5-d)	53.3% binary acc.	
Linear (frozen MERT)	MERT 768-d	34.4	56.4
KNN (k=1, frozen MERT)	MERT 768-d	20.4	20.7
KNN (k=5, frozen MERT)	MERT 768-d	16.9	33.8
MLP search (unperturbed)	MERT 768-d	38.4	59.1
MLP search (perturbed)†	MERT 768-d	87.8	96.6
MLP search (leak-corrected)	MERT 768-d	44.9	69.3
Fine-tune MERT (2 layers)	Raw audio	68.9	87.9
Fine-tune MERT (4 layers)	Raw audio	**70.2**	**88.1**
Metric learning (ArcFace)	MERT 768-d	40.2	62.9

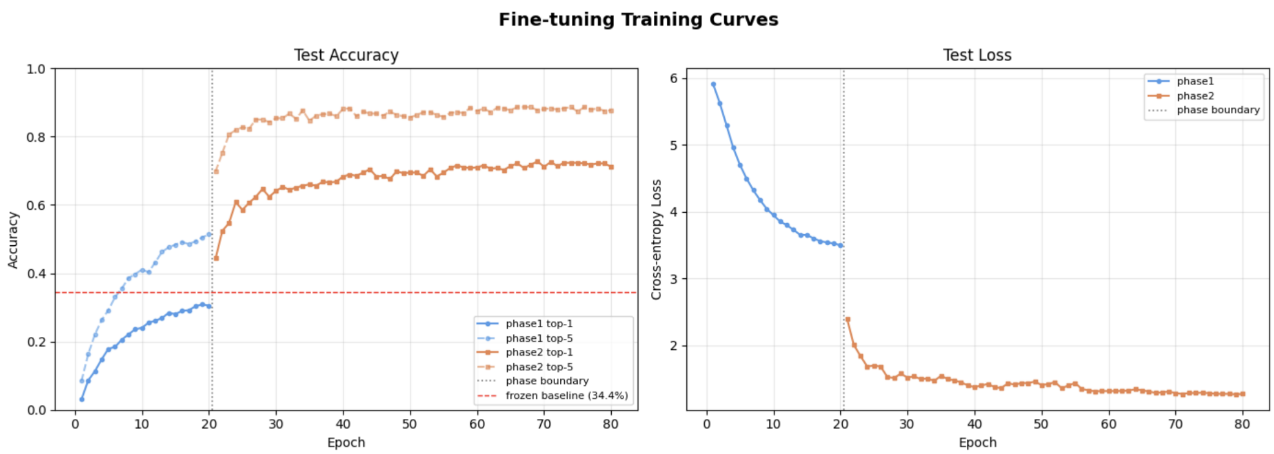

Fig. 1. Fine-tuning performance curves on the unperturbed dataset. During Phase 1, the MERT backbone is frozen and only the classification head is trained. During Phase 2, the upper MERT transformer layers are unfrozen and optimized end-to-end. Accuracy rises sharply after unfreezing, supporting the conclusion that adapting the embedding representation is more effective than increasing downstream classifier capacity alone. The dashed horizontal line marks the frozen linear-classifier baseline of 34.4% top-1 accuracy. The curves are used to visualize optimization behavior rather than to select model checkpoints.

(see Section IV-C). The inflated metrics thus reflect near-memorisation of perturbation variants rather than genuine piece identification. After correcting the split to assign all perturbations of a given snippet exclusively to one partition, the valid result was 44.9% top-1 / 69.3% top-5 — a meaningful but modest improvement over the unperturbed MLP search (38.4%). The leakage incident is recorded here as a methodological observation: it illustrates both the sensitivity of evaluation to split design and the degree to which snippet-level contamination can inflate reported accuracy.

d) Fine-Tuning MERT.: Fine-tuning the MERT backbone itself proved to be the decisive intervention. As shown in Fig. 1, performance improves sharply after the phase boundary, when the upper MERT layers are unfrozen. This supports the interpretation that the main bottleneck was the frozen embedding geometry rather than the capacity of the downstream classifier. Using a two-phase strategy — a 20-epoch warm-up of the linear head on frozen embeddings, followed by end-to-end training with differential learning rates (10^{-5} for the transformer layers, 10^{-3} for the head) — the model with two unfrozen transformer layers reached 68.9% top-1 / 87.9% top-5, a 2.00× improvement over the frozen linear baseline. Unfreezing four layers improved further to **70.2%** top-1 / **88.1%** top-5, the best result obtained in this work. Both fine-tuning runs converged without catastrophic forgetting; the four-layer model's test loss was still declining at epoch 30, suggesting that additional training would yield further gains within the compute budget available.

e) Metric Learning.: Applying an ArcFace classification head on top of a projection network over frozen MERT embeddings achieved 40.2% top-1 / 62.9% top-5. This result modestly exceeds the frozen linear classifier (34.4%) but falls approximately 30 percentage points below fine-tuned MERT and slightly below the leak-corrected augmented MLP (44.9%). The limited gain from a more discriminative loss function, relative to the large gain from fine-tuning, supports the conclusion that the primary bottleneck throughout the frozen-embedding regime was the representational geometry of MERT rather than the choice of classification objective.

VII. Discussion

The results across all eight methods point to a single unifying diagnostic: throughout the project, the primary bottleneck was the geometry of the embedding space, not the capacity of the classifier sitting on top of it.

The most direct evidence for this claim comes from the plateau shared by every frozen-embedding method. A logistic regression over MERT embeddings reached 34.4% top-1 accuracy; an MLP with a 512-unit hidden layer, cosine annealing, label smoothing, and inverse-frequency class weighting reached 38.4% on the unperturbed dataset, while leak-corrected pitch-and-tempo augmentation improved the best frozen result to 44.9%. The ceiling moved only modestly despite a substantial increase in model capacity and training sophistication. Regularization sweeps ruled out overfitting as the limiting factor — weight-decay variation across three orders of magnitude changed test accuracy by less than 0.6 percentage points. The correct interpretation is that the frozen MERT embeddings do not arrange the 430 Bach chorales in a linearly (or even simply nonlinearly) separable configuration; the most the classifier can do is identify the small fraction of pieces whose embeddings happen to be geometrically isolated.

Together, the experiments show that frozen MERT embeddings contain substantial piece-identity information, but their raw geometry is not well aligned with fine-grained chorale classification. Downstream classifiers improve performance only modestly, while fine-tuning the upper transformer layers substantially reorganizes the representation space and produces the strongest results.

Fine-tuning the MERT transformer itself resolved this. By adapting the upper layers of the model end-to-end on the piece-identity objective, the embedding space was reorganized around chorale structure rather than general audio properties. Both fine-tuning configurations produced an immediate doubling of top-1 accuracy at the very first epoch of Phase 2, confirming that the representation was the constraint, not the head. The 4-layer run ultimately reached 70.2% top-1 and 88.1% top-5 — a $2.04\times$ improvement over the frozen baseline — while the loss curve was still declining at epoch 30, indicating that the gains are not yet saturated.

The metric learning results reinforce this geometry-first diagnosis from the opposite direction. Metric learning explicitly optimizes embedding geometry via a projection head and cosine classifier, yet it underperformed fine-tuning by approximately 30 percentage points in top-1 accuracy. The reason is instructive: metric learning reshapes the space only through the shallow projection head, leaving the MERT backbone frozen. Fine-tuning, by contrast, adjusts the representations at their source. When the backbone is frozen, no downstream head — whether linear, nonlinear, or geometry-optimizing — can fully compensate for a space not organized around the target task.

Finally, the data augmentation episode offers a practical lesson in experimental design. Pitch shifting and tempo stretching expanded the effective training set by a factor of 25 and genuinely improved robustness to acoustic variation. However, the inflated accuracy figures produced by the leaky train/test split reveal how easily augmentation can confound evaluation: snippets and their perturbations must be separated at the snippet level, not the piece level, to prevent the model from learning to recognize acoustic twins rather than generalizing across pieces. Correcting the split reduced reported accuracy to the true performance level, which remained the highest among frozen-embedding methods and underscored that augmentation adds real value once the evaluation is made rigorous.

VIII. Limitations

Several scope constraints bound the conclusions that can be drawn from this work.

The most significant is the dataset's homogeneity. All 430 pieces are Bach chorales drawn from a single corpus, sharing a common era, harmonic language, instrumentation, and dynamic range. This uniformity likely makes the classification task easier than a real-world scenario involving mixed composers, genres, or recording conditions, where inter-class similarity is far lower and intra-class variation is far higher. Whether the fine-tuned MERT representations transfer to jazz, symphonic, or multi-instrument repertoire is an open question.

Compute availability imposed a ceiling on the fine-tuning experiments. Both MERT runs were conducted on Colab GPU instances with a practical limit of 30 Phase 2 epochs. The 4-layer run's loss curve was still declining at the final checkpoint, suggesting that continued training would yield further accuracy gains. The reported 70.2% top-1 result should therefore be understood as a lower bound on what fine-tuning can achieve, not a convergence point.

A structural limitation is the mean-pooling step used to compress MERT's per-frame hidden states into a single 768-dimensional vector. Mean-pooling discards the temporal ordering of frames, reducing an 8-second sequence to a bag of activations. Musical identity is partly a sequential property — motifs, rhythmic patterns, and harmonic progressions unfold over time — and a classifier that operates on a pooled summary cannot exploit this structure. A sequence-aware aggregation method could recover information lost in this step.

The metric learning results are preliminary. Only one configuration was evaluated, ablations over projection head depth, margin parameters, and loss formulation were not completed, and the comparison to fine-tuning is therefore not fully controlled. More systematic experimentation is needed before drawing strong conclusions about the relative merits of the two approaches.

Finally, class imbalance is a persistent challenge. Individual chorales produce between roughly 3 and 40 training snippets depending on piece length. The inverse-frequency weighting applied in training mitigates but does not eliminate the effect; short pieces remain systematically harder to classify, and this imbalance may skew aggregate accuracy figures toward longer works.

IX. Future Work

The findings in this paper open several natural directions for future investigation.

The most immediate extension is dataset broadening. The current work establishes that fine-tuned MERT representations can solve fine-grained classification within a homogeneous single-composer corpus. Whether this holds across composers, genres, and eras — Bach alongside jazz standards, classical symphonies, and folk recordings — would test the generality of the approach and better approximate the demands of real-world music identification systems. A cross-composer benchmark would also introduce harder inter-class confusion, providing a more demanding evaluation of embedding quality.

Within the existing architecture, replacing mean-pooling with a sequence-aware aggregation head is a high-priority improvement. MERT produces a full sequence of per-frame hidden states, and there is no reason to believe that a simple average captures the temporal structure that distinguishes one chorale from another. An attention-based pooling layer or a lightweight recurrent head trained over the frame sequence could exploit motif repetition, harmonic rhythm, and cadential patterns that the current model discards.

On the representation learning side, end-to-end metric learning trained jointly with the MERT backbone is the natural evolution beyond the preliminary results reported here. Triplet loss or a CLAP-style contrastive objective applied directly to audio would optimize the embedding space for piece identity from the first training step rather than as a post-hoc projection. Combined with full fine-tuning of the backbone, this approach could yield representations more tightly clustered by piece than classification-based fine-tuning produces.

A complementary direction is augmentation in embedding space rather than at the waveform level. Perturbing the 768-dimensional representations directly — for example, by adding noise in directions of low intra-piece variance — would be computationally cheaper than re-running FluidSynth and MERT inference on augmented audio, and might produce more targeted regularization.

Finally, a systematic misclassification analysis would both diagnose remaining failure modes and inform data collection priorities. Current evidence suggests that pieces are most often confused along key and harmonic similarity lines; understanding whether classification error correlates with piece length, snippet count, or harmonic proximity to neighboring chorales would clarify which aspects of the representation are still underspecified and where additional training signal would be most valuable. The broader applicability to cover song identification — where the same musical material appears in different arrangements — makes this analysis relevant beyond the Bach chorale setting.

X. Conclusion

This paper addressed fine-grained music piece identification: classifying 8-second audio snippets to their source piece among 430 Bach chorales, against a random baseline of 0.23% top-1 / 1.16% top-5. We traced a full methodological arc from hand-crafted feature-based logistic regression through frozen transformer embeddings with MLP classifiers, ultimately to task-specific fine-tuning of the MERT audio transformer. The central finding is that embedding geometry, not classifier capacity, determines classification performance: frozen-embedding methods plateaued in the 38–45% top-1 range, while fine-tuning the top four MERT transformer layers broke through this ceiling to **70.2% top-1 / 88.1% top-5** accuracy — a 305$\times$ improvement over random chance. These results establish that pre-trained audio transformers can be effectively specialized for fine-grained within-corpus identification through targeted fine-tuning, with potential applicability to music recommendation, cover song detection, and large-scale archival search.

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